

AWS Certified AI Practitioner AIF-C01

Section 10: Amazon Sagemaker

A) What is Amazon SageMaker?

1. The basic idea

Amazon SageMaker AI is a fully managed AWS service for building, training, tuning, deploying, and managing machine learning models.

Imagine a company wants to predict whether customers will cancel their subscriptions. The company has historical customer data such as:

- Customer age
- Subscription length
- Monthly spending
- Number of support calls
- Number of logins
- Whether the customer eventually cancelled

A machine learning team could use this historical data to train a model that predicts whether a current customer is likely to cancel.

For the exam, remember:

SageMaker = AWS's managed platform for the machine learning lifecycle.

2. What does "fully managed" mean?

Without SageMaker, an organization might have to manually configure servers, install machine-learning frameworks, manage training infrastructure, configure GPUs, build model-serving infrastructure, scale endpoints, monitor models, and maintain other ML infrastructure.

With SageMaker, AWS manages much of the underlying infrastructure.

That does **not** mean AWS automatically creates the perfect model. Data scientists still need to make decisions about data, algorithms, features, evaluation, security, deployment, and monitoring.

"Fully managed" mainly means AWS reduces the infrastructure-management burden.

3. Training

Training means allowing the ML algorithm to learn patterns from historical data.

Conceptually:

Training Data



ML Algorithm
↓
Learning
↓
Trained Model

After training, the model can be given previously unseen data and asked to make predictions.

For example:

New transaction
Amount = \$9,000
Device = New
Location = Foreign
Time = 2:30 AM
↓
Trained Model
↓
Fraud probability = 94%

That second process is called **inference**.

Exam distinction

Training = model learns.

Inference = trained model makes predictions.

4. Hyperparameter Tuning

A model contains **parameters** and **hyperparameters**.

Parameters are generally learned during training.

Hyperparameters are configuration values that influence the training process.

Examples include:

- Learning rate
- Batch size
- Number of trees
- Number of training epochs
- Regularization settings

Instead of a data scientist manually trying many combinations, **SageMaker Automatic Model Tuning** can run training jobs with different hyperparameter combinations and search for configurations that optimize a selected metric.

Conceptually:

Configuration A → Accuracy 87%
Configuration B → Accuracy 91%
Configuration C → Accuracy 89%
↑
Better configuration

Exam clue

If AWS says:

"Automatically find the best hyperparameter values"

"The data scientist wants AWS to automatically search for the best hyperparameter configuration."

Think:

SageMaker Automatic Model Tuning / Hyperparameter Optimization.

5. Deploying the Model

Once a model has been trained, applications need to use it.

This is **inference**.

SageMaker provides several inference options. Four particularly important ones for the AI Practitioner exam are:

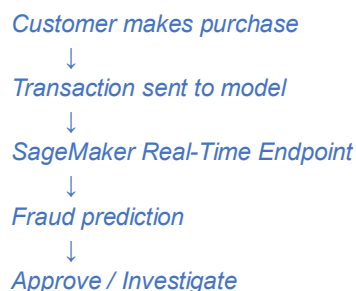
1. **Real-Time Inference**
2. **Serverless Inference**
3. **Asynchronous Inference**
4. **Batch Transform**

The differences are highly testable.

a) Real-Time Inference

Real-Time Inference is designed for interactive, low-latency predictions.

Suppose a bank needs to evaluate every credit-card transaction while the customer is making the purchase.



The prediction needs to happen quickly.

Common examples include:

- Fraud detection
- Real-time recommendations

- Risk scoring
- Live personalization
- Interactive ML applications

Exam keywords

If you see:

"low latency" + "immediate prediction" + "real-time application"

think:

Real-Time Inference

b) Serverless Inference

Serverless Inference is useful when traffic is **intermittent or unpredictable** and you don't want to manage the underlying inference infrastructure.

SageMaker automatically manages compute capacity.

You pay for the compute used to process inference requests rather than maintaining idle provisioned capacity. AWS specifically positions this option for intermittent/unpredictable traffic.

Consider an internal HR application.

Employees occasionally submit a document for classification:

8:00 AM 0 requests
 9:00 AM 2 requests
 10:00 AM 0 requests
 11:00 AM 1 request

Maintaining a continuously provisioned endpoint could be unnecessary.

Serverless inference is a better fit.

Exam clue

Look for:

"Unpredictable traffic"
 "Intermittent traffic"
 "Idle periods"
 "Don't want to manage inference infrastructure"

Think **Serverless Inference**.

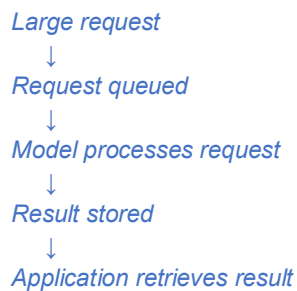
c) Asynchronous Inference

Asynchronous Inference is for requests that take longer to process, may involve large payloads, and don't require the client to wait synchronously for an immediate response.

Requests are queued and processed asynchronously.

AWS currently supports payloads up to **1 GB** and processing times up to **one hour** for this inference option.

Conceptually:



Examples

Large:

- Images
- Documents
- Audio
- Video-related inputs

Exam clue

"Large payload + long processing time + request can be queued"

Think **Asynchronous Inference**.

d) Batch Transform

Suppose a retailer wants predictions for **10 million customers every Sunday night**.

The predictions don't need to happen interactively.

The company already has a dataset containing all 10 million records.

It can process the entire dataset as a batch.

10 million records



Batch Transform



ML Model



10 million predictions



Results

This is **Batch Transform**.

It's appropriate for offline predictions over large datasets where a persistent endpoint isn't required.

Exam keywords

If you see:

- Large dataset
- Offline predictions
- Scheduled predictions
- Process millions of records
- No persistent endpoint

think:

Batch Transform

Real-Time vs Serverless vs Async vs Batch

This table is worth memorizing.

Scenario	Choose
Immediate prediction	Real-Time
Low latency	Real-Time
Sustained interactive traffic	Real-Time
Sporadic traffic	Serverless
Unpredictable traffic	Serverless
Avoid idle provisioned inference capacity	Serverless
Long-running individual request	Asynchronous
Large payload	Asynchronous
Request should be queued	Asynchronous
Millions of records together	Batch Transform
Offline processing	Batch Transform
Scheduled bulk predictions	Batch Transform

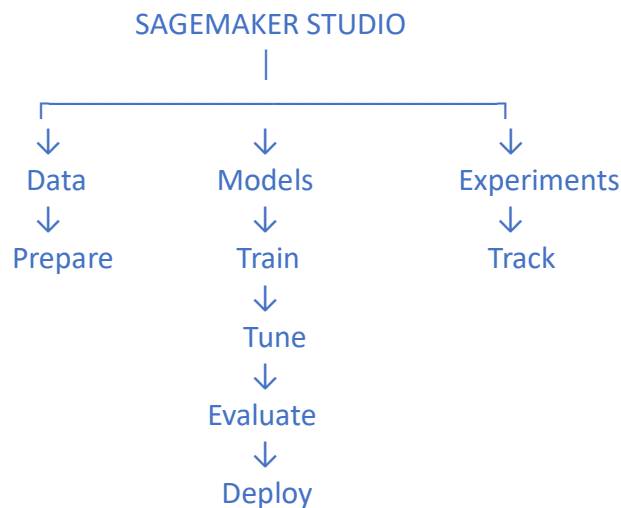
A useful mental shortcut:

Real-Time = NOW
Serverless = SOMETIMES
Async = LATER
Batch = MANY

6. SageMaker Studio

SageMaker Studio is an integrated development environment for machine learning.

Think of it as a central workspace where data scientists and ML engineers can access tools needed throughout the ML lifecycle.



SageMaker AI = managed ML platform/service.

SageMaker Studio = integrated ML development environment.

B) Amazon SageMaker Data Tools

Now we move backward in the ML lifecycle.

Before training a model, you need good data.

Two important concepts are:

Data Wrangler → prepare the data

Feature Store → store and reuse ML features

1. Why Data Preparation Matters

Suppose you receive this dataset:

Customer	Age	Salary	City
John	35	\$75,000	Toronto
Mary	Missing	\$82,000	Vancouver
Alex	29	Missing	TOR
Sarah	41	\$105,000	Toronto

There are problems:

- Missing values

- Inconsistent city names
- Potentially irrelevant columns
- Different data formats

If you train a model directly on poor-quality data, the resulting model may perform poorly.

This leads to an important ML principle:

Good ML models depend heavily on good data.

2. SageMaker Data Wrangler

SageMaker Data Wrangler helps prepare and transform data for machine learning.

It can help with tasks such as:

- Importing data
- Cleaning data
- Handling missing values
- Transforming columns
- Encoding categorical variables
- Normalization
- Detecting anomalies
- Feature engineering
- Visualizing data
- Creating reusable data preparation flows

For the AI Practitioner exam:

Data Wrangler = data preparation and transformation.

3. Feature Engineering

A **feature** is an input used by an ML model.

For example:

Age

Income

Credit Score

Account Balance

are potential features for a financial ML model.

Feature engineering means creating or transforming features to make them more useful.

Suppose the original data contains:

Date of Birth = 1990-05-10

You could derive:

Age = 36

That's feature engineering.

Or suppose you have:

Annual Spending = \$20,000

Annual Income = \$100,000

You could create:

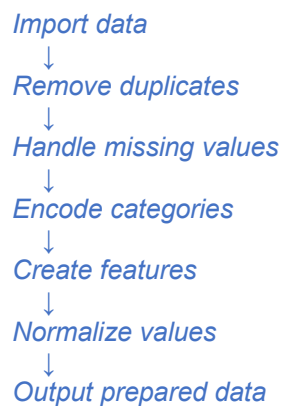
Spending Ratio = 0.20

The newly created Spending Ratio can become another feature.

4. Data Wrangler Data Flows

A **data flow** records a sequence of data transformations.

For example:



This makes the preparation process more reproducible.

Instead of manually performing the transformations every time, the flow captures the sequence.

5. SageMaker Feature Store

Now suppose the company creates a useful feature:

Customer Risk Score

The fraud team needs it.

The lending team needs it.

The customer-retention team needs it.

Without a centralized feature repository, each team could independently recreate the same feature.

That creates duplication and potentially inconsistent calculations.

SageMaker Feature Store provides centralized storage and management for machine-learning features.

Think:

FEATURE STORE

|



All three models can use standardized features.

Exam clue

"Multiple ML teams need to reuse the same engineered features."

Think:

SageMaker Feature Store

a. Online Feature Store

The **online store** is designed for low-latency retrieval of current feature values.

Imagine a live fraud transaction.

The fraud model needs:

Current customer risk score

Current spending pattern

Recent transaction count

immediately.

Use the:

Online Feature Store

Think:

Online = live/low-latency feature retrieval

b. Offline Feature Store

The **offline store** is appropriate for historical feature data used for activities such as training and batch analysis.

For example:

Three years of historical

customer risk features



Offline Feature Store



Train new fraud model

Think:

Offline = historical/training/batch

6. Data Wrangler vs Feature Store

This is an important exam distinction.

Data Wrangler

Answers:

How do I prepare and transform this data?

Feature Store

Answers:

Where do I centrally store and reuse these ML features?

So:



C) SageMaker — Models and Humans

Now we move into **model evaluation, responsible AI and human involvement**.

Four concepts are particularly important:

1. SageMaker Clarify
2. SageMaker Ground Truth
3. Human evaluation
4. RLHF

1. SageMaker Clarify

SageMaker Clarify helps with **bias detection and model explainability**.

Imagine a bank creates a loan-approval model.

The model considers:

Income

Credit Score

Debt

Employment History

and outputs:

Approved

or

Rejected

The bank should investigate two questions:

Question 1 — Bias

Is the model producing systematically different outcomes for particular groups in ways the organization needs to examine?

Question 2 — Explainability

Why did the model produce a particular prediction?

SageMaker Clarify helps with these kinds of analyses.

For the exam:

Clarify = Bias + Explainability

a. Bias

ML models learn from data.

If training data contains problematic historical patterns, imbalances, or representation issues, a model may reproduce or amplify them.

Suppose a company trains a hiring model on historical hiring data.

If the historical dataset disproportionately represents one demographic group, the organization needs to evaluate whether undesirable bias is present.

Clarify provides bias metrics and analysis to help identify such issues.

Exam scenario

"A financial institution wants to evaluate whether its ML model contains potential bias."

Answer:

SageMaker Clarify

b. Explainability

Some ML models are difficult to interpret.

Imagine a model says:

Loan application → REJECTED

The customer or business may reasonably ask:

Why?

Explainability attempts to identify which features influenced the prediction.

For example:

High debt → strong influence

Low credit score → strong influence

Income → moderate influence

Employment duration → smaller influence

Clarify can help provide model explainability.

Exam clue

"Which features influenced the prediction?"

Think:

SageMaker Clarify

2. SageMaker Ground Truth

SageMaker Ground Truth is primarily associated with creating **labeled training data**.

Imagine you have:

500,000 images

but none are labeled.

A computer-vision model needs examples such as:

Image 1 → Cat

Image 2 → Dog

Image 3 → Car

Image 4 → Bicycle

Humans can annotate these images.

Ground Truth helps manage data-labeling workflows.

Other examples include:

Text classification

"I love this product" → Positive

Object detection

Image → bounding box around car

Audio

Audio → transcription/label

Exam memory

Ground Truth = Label the data

3. Human-in-the-Loop

Human-in-the-loop (HITL) means humans remain involved in an AI/ML process rather than allowing the system to make every decision automatically.

Imagine an insurance-claim model.

High-confidence prediction:

AI confidence = 99%

↓

Automated workflow

Low-confidence prediction:

AI confidence = 55%

↓

Human reviewer

↓
Final decision

Human involvement can be particularly important for:

- Sensitive decisions
- Ambiguous cases
- Low-confidence predictions
- Quality assurance
- Regulatory or organizational requirements

Evaluating Foundation Models

Suppose a company is choosing between two foundation models.

It asks both:

"Summarize this customer complaint."

Model A produces one response.

Model B produces another.

The company could compare them based on dimensions such as:

- Accuracy
- Relevance
- Helpfulness
- Coherence
- Safety
- Bias
- Latency
- Cost

Not every desirable property can be adequately measured by a simple automatic metric.

That's where **human evaluation** can be useful.

Humans can compare model outputs and express preferences.

4. RLHF

RLHF = Reinforcement Learning from Human Feedback.

Human preferences can also be used to help align model behavior.

Simplified:

User Prompt

↓

Model generates responses

↓

Humans rank/prefer responses

↓
Human preference signal
↓
Training/alignment process

↓
Improved model behavior

Example:

Prompt:

"Explain cloud computing to a beginner."

Model produces Response A and Response B.

Humans repeatedly indicate which responses are more useful or appropriate.

Those preferences can contribute to training/alignment.

Human Evaluation vs RLHF

They are related but not identical.

Human evaluation

Humans judge model outputs.

Purpose:

Evaluate/compare model quality.

RLHF

Human preference information contributes to model training/alignment.

Purpose:

Improve/align model behavior.

Remember:

Human Evaluation = Judge

RLHF = Learn from judgments

D) SageMaker Governance

Building a model isn't the end of the ML lifecycle.

Organizations also need to control, document, monitor and manage models.

Suppose a bank has **200 ML models**.

It needs to know:

- Who created each model?
- What is each model used for?
- Which version is in production?
- Which models have been approved?

- Are models still performing correctly?
- Are they experiencing drift?
- Who has permission to manage them?

This is **ML governance**.

AWS groups SageMaker Model Cards, Model Dashboard and Role Manager among its purpose-built governance capabilities.

1. SageMaker Model Cards

Model Cards provide structured documentation about ML models.

A Model Card can document information such as:

- Model purpose
- Intended use
- Risk information
- Training details
- Evaluation information
- Model characteristics

Think of it as a **structured information record/documentation for a model**.

Example

MODEL CARD

Model:

Loan Risk Predictor v4

Purpose:

Predict loan default risk

Intended use:

Consumer loan assessment

Evaluation:

Accuracy, precision, recall...

Risk considerations:

Bias and fairness considerations

Exam clue

"Company needs standardized documentation describing a model's purpose, evaluation and risk."

Think:

Model Cards

2. Model Dashboard

When an organization has many models, it needs centralized visibility.

SageMaker Model Dashboard provides a centralized view for monitoring models and related information.

Think of it as:

A control/visibility dashboard for models.

It can help surface information related to:

- Models
- Endpoints
- Model performance
- Data quality
- Model quality
- Bias
- Explainability
- Monitoring issues

Exam clue

"Centralized view of deployed models and monitoring information"

Think:

Model Dashboard

3. Role Manager

ML teams contain different personas.

For example:

Data Scientist

ML Engineer

MLOps Engineer

Administrator

They should not necessarily have identical AWS permissions.

SageMaker Role Manager helps administrators create and manage persona-based roles and permissions for ML activities.

This supports the security principle of:

Least privilege

Meaning:

Give users only the permissions they need.

Exam clue

"Administrator wants persona-based least-privilege permissions for ML users."

Think:

Role Manager

4. Model Registry

Models evolve.

Suppose:

Fraud Model v1

Fraud Model v2

Fraud Model v3

Fraud Model v4

Which one should be deployed?

Which one has been approved?

Which one is waiting for evaluation?

SageMaker Model Registry helps catalog model versions and manage them through their lifecycle.

Think:

Model Registry = model version/lifecycle management

Model Cards vs Model Registry

This is a common conceptual trap.

Model Cards

Document the model.

Model Registry

Manage model versions/lifecycle.

Example:

Fraud Model v5

|

/— Model Card

|

└— What is this model?

|

└— Model Registry

└— Which version/state is it in?

5. SageMaker Pipelines

ML workflows contain multiple steps.

For example:

Prepare data



Train model



Evaluate model



Quality check



Register model

↓
Deploy

Doing this manually repeatedly is inefficient and error-prone.

SageMaker Pipelines provides workflow orchestration for machine-learning pipelines.

It helps make ML processes:

- Automated
- Repeatable
- Consistent
- Easier to manage

Exam clue

"Automate an end-to-end ML workflow."

Think:

SageMaker Pipelines

E) SageMaker Consoles and Related Tools

Now we come to:

- JumpStart
- Canvas
- MLflow

These solve very different problems.

1. SageMaker JumpStart

Imagine a company wants to build an ML solution.

One option is:

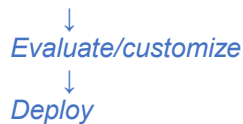
Collect huge dataset
↓
Design model
↓
Train from scratch
↓
Evaluate
↓
Deploy

But pretrained models already exist for many tasks.

SageMaker JumpStart acts as an ML hub that provides access to pretrained models and other ML resources so teams can get started more quickly.

Conceptually:

SageMaker JumpStart
↓
Find pretrained model



For the exam:

JumpStart = ML hub + pretrained models + faster ML development

Pretrained Models

A **pretrained model** has already been trained on substantial data.

Instead of starting from zero, organizations can start with an existing model and use, customize, or fine-tune it where appropriate.

This can save:

- Training time
- Compute
- Development effort
- Data requirements

Model ecosystems/providers can include organizations such as **Hugging Face** and **Meta**, depending on available offerings.

For AI Practitioner, don't spend much time memorizing individual models.

Understand the principle:

Pretrained model = don't necessarily train from scratch.

2. SageMaker Canvas

SageMaker Canvas provides a visual, low-code/no-code experience for working with data and machine learning.

Imagine a business analyst understands the company's data but doesn't know Python or TensorFlow.

The analyst still wants to:

- Prepare data
- Explore data
- Build ML models
- Generate predictions

Canvas provides a visual interface that lowers the programming barrier.

Exam clue

"Business analyst has little ML programming experience and needs a visual ML interface."

Think:

SageMaker Canvas

3. MLflow

Data scientists frequently run many experiments.

For example:

Experiment 1
Model = XGBoost
Accuracy = 87%

Experiment 2
Model = XGBoost
Accuracy = 91%
Different hyperparameters

Experiment 3
Model = Neural Network
Accuracy = 89%

Soon they need answers to:

- Which experiment performed best?
- Which parameters did we use?
- What metrics did it produce?
- What artifacts belong to the run?
- Can another team member reproduce the experiment?

MLflow is an open-source platform associated with managing the ML lifecycle and particularly experiment tracking.

For the exam, think:

MLflow = track ML experiments, parameters, metrics and artifacts.

JumpStart vs Canvas vs Studio vs MLflow

These are easy to confuse.

Tool	Think
SageMaker Studio	Integrated ML development environment
SageMaker JumpStart	ML hub/pretrained models
SageMaker Canvas	Visual/no-code/low-code ML
MLflow	Experiment tracking/lifecycle management

Memory shortcut:

Studio → Develop

JumpStart → Start quickly

Canvas → No/low code

MLflow → Track experiments

AWS AI Practitioner Scenario-Based Questions

Try answering these **before reading the answers**.

Question 1

A financial company needs to evaluate every credit-card transaction for fraud while the transaction is taking place. Predictions must have low latency.

Which option should it use?

- A. Batch Transform
- B. Asynchronous Inference
- C. Real-Time Inference
- D. Data Wrangler

Answer: C — Real-Time Inference

Why: Immediate, low-latency predictions are required.

Question 2

A company's ML application receives prediction requests only a few times each day. Traffic is unpredictable, and the company wants to avoid maintaining unnecessary inference capacity.

- A. Real-Time Inference
- B. Serverless Inference
- C. Batch Transform
- D. Feature Store

Answer: B — Serverless Inference

Exam trigger: intermittent/unpredictable traffic.

Question 3

A company processes large individual files with an ML model. Each request can take several minutes, and the request can be queued.

- A. Real-Time
- B. Serverless
- C. Asynchronous
- D. Batch

Answer: C — Asynchronous Inference

Remember:

Large + long + queued → Async

Question 4

A retailer wants to calculate churn predictions for 15 million customers every Saturday night.

- A. Real-Time
- B. Batch Transform

- C. Serverless
- D. Ground Truth

Answer: B — Batch Transform

Because this is a **large offline dataset**.

Question 5

A dataset contains missing values, inconsistent formats and categorical variables that need transformation before training.

- A. Data Wrangler
- B. Feature Store
- C. Model Registry
- D. Clarify

Answer: A — Data Wrangler

Question 6

Three ML teams want to reuse the same engineered customer-risk features.

- A. Data Wrangler
- B. Feature Store
- C. Ground Truth
- D. Model Dashboard

Answer: B — Feature Store

The key word is **reuse**.

Question 7

A fraud model requires the customer's latest risk score with very low latency during a transaction.

- A. Offline Feature Store
- B. Online Feature Store
- C. Data Wrangler
- D. Batch Transform

Answer: B — Online Feature Store

Online → current + low latency

Question 8

A company needs three years of historical feature data to train a new model.

- A. Online Feature Store
- B. Offline Feature Store
- C. Real-Time Endpoint
- D. Clarify

Answer: B — Offline Feature Store

Offline → historical/training

Question 9

A bank wants to determine whether its loan model exhibits potential bias.

- A. Ground Truth
- B. Clarify

- C. JumpStart
- D. Canvas

Answer: B — Clarify

Question 10

A bank wants to understand which features contributed to a loan model's prediction.

- A. Clarify
- B. Ground Truth
- C. MLflow
- D. Feature Store

Answer: A — Clarify

Because the requirement is **explainability**.

Question 11

A company has 400,000 images that humans need to annotate before the images can be used for training.

- A. Clarify
- B. Ground Truth
- C. Model Cards
- D. Pipelines

Answer: B — Ground Truth

Because the requirement is **data labeling**.

Question 12

A company asks humans to compare responses from two foundation models based on helpfulness and relevance.

What is being performed?

- A. Feature engineering
- B. Human evaluation
- C. Batch inference
- D. Model drift

Answer: B — Human evaluation

Question 13

Human preferences about generated responses are subsequently used as part of a reinforcement-learning process to align a model.

What is this?

- A. RAG
- B. RLHF
- C. Feature engineering
- D. Batch Transform

Answer: B — RLHF

Question 14

A compliance team needs documentation describing a model's intended use, risk and evaluation information.

- A. Model Registry
- B. Model Cards
- C. Model Dashboard
- D. Feature Store

Answer: B — Model Cards

Question 15

A company wants a centralized view of models and their monitoring information.

- A. Model Dashboard
- B. Ground Truth
- C. JumpStart
- D. Data Wrangler

Answer: A — Model Dashboard

Question 16

An administrator needs to create persona-based, least-privilege permissions for data scientists and ML engineers.

- A. Role Manager
- B. Clarify
- C. MLflow
- D. Model Cards

Answer: A — Role Manager

Question 17

An ML team has versions 1 through 15 of a fraud model and needs to manage model versions and lifecycle states.

- A. Model Cards
- B. Model Registry
- C. Data Wrangler
- D. Ground Truth

Answer: B — Model Registry

Question 18

A company wants to automate:

Prepare → Train → Evaluate → Register

Which capability should it use?

- A. MLflow
- B. SageMaker Pipelines
- C. Clarify
- D. Ground Truth

Answer: B — SageMaker Pipelines

Question 19

A recommendation model has become less effective because customer purchasing behavior has changed considerably since training.

What should the team investigate?

- A. Model/data drift
- B. Ground Truth
- C. Serverless cold start
- D. Hyperparameter initialization

Answer: A — Model/data drift

Question 20

A company wants to quickly begin an ML project using pretrained models instead of developing and training everything from scratch.

- A. SageMaker JumpStart
- B. SageMaker Clarify
- C. Model Dashboard
- D. Data Wrangler

Answer: A — SageMaker JumpStart

Question 21

A business analyst has little programming experience but wants a visual interface for working with ML.

- A. MLflow
- B. SageMaker Canvas
- C. Clarify
- D. Model Registry

Answer: B — SageMaker Canvas

Question 22

Data scientists want to track experiments, parameters, metrics and artifacts across different ML runs.

- A. MLflow
- B. Ground Truth
- C. Data Wrangler
- D. Role Manager

Answer: A — MLflow